

PBA: 4. Causality

Fall 2026

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Where are we? Where are we going?

- **Where we have been**

- Week 1: R and RStudio, functions and packages, Quarto
- Week 2: data visualization. A table became a graph, layer by layer, with `ggplot2`
- Week 3: data wrangling with `dplyr`. Rows, columns, and groups, then barplots from the summaries

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- **Today: causality**

- What is a causal effect? Randomized experiments, calculating effects, and observational studies
- Building on last week: `filter()`, `group_by()`, and `summarize()` now compare a treated group with a control group
- After the lecture you run today's code yourself, then E3 at your own pace with the TA

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- **Where we are going**

- Next week (Oct 8): causality II. DAGs, and what to control for in your project. Then prediction, regression, and uncertainty

Roadmap

1. What is causality?
2. Randomized experiments
3. Calculating effects
4. Observational studies
5. In-Class Exercise E3

Before we begin..
Rising Interest in 'Causal' Data Science



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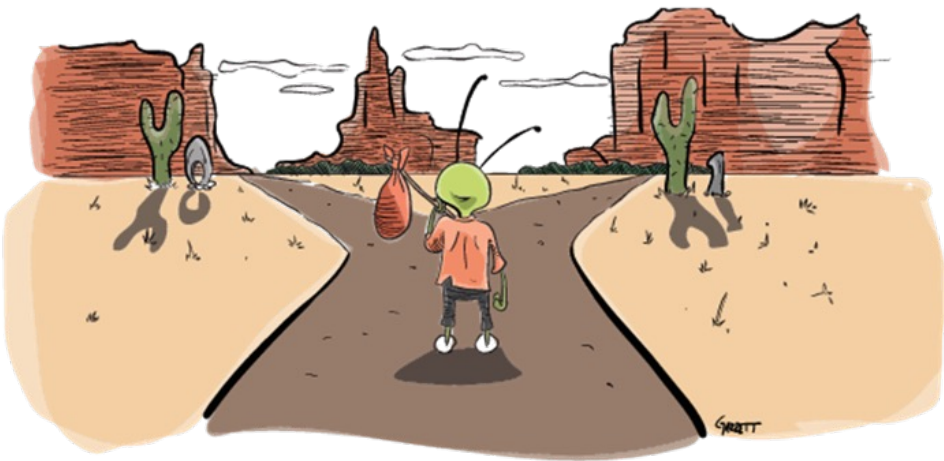
1/ What is causality?



The Road Not Taken:

*"Two roads diverged in a yellow wood,
And sorry I could not travel both
And be one traveler, long I stood
And looked down one as far as I could
To where it bent in the undergrowth;"*

– Robert Frost



Source: *Chapter 1 of Mastering Metrics* by J. Angrist & J. Pischke

What is a Causal Effect?

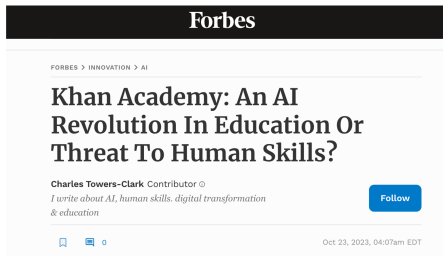
Factual

vs.

Counterfactual

- Does increasing the minimum wage increase the unemployment rate?
 1. Unemployment rate went up after the minimum wage increased.
 2. Would the rate have risen even without the wage increase?
- Does having a CEO with a hobby of flying airplanes lead to better innovation outcomes? ([here](#))
 1. Innovation outcomes are higher for companies led by pilot CEOs.
 2. Would the innovation outcomes have been the same w/o a pilot CEO?
- **Fundamental problem of causal inference:**
 - ↪ Can never observe counterfactuals, must be inferred!

Context: AI Tutoring Study



<https://bit.ly/45VfES8>



Created using DALLE-3

- Can AI assistants improve students' learning outcomes?
- Experimental setting:
 - Randomly assign students to have tutoring sessions with or without AI assistants.
 - AI-assisted sessions focus on adaptive learning techniques.
- Outcome of interest:
 - Improvement in test scores or grades.

A Tale of Two Students

	AI Assistant	Test Result
Student 1	Tutoring without AI assistant	Fail
Student 2	Tutoring with AI assistant	Pass

- Q: Did the second respondent pass the standardized exam **because** of the AI assistant?

Translating into Math

- Useful to have **compact** notation for referring to **treatment variable**:

$$T_i = \begin{cases} 1 & \text{if student } i \text{ had access to tutoring with AI assistant} \\ 0 & \text{if student } i \text{ received tutoring without AI assistant} \end{cases}$$

- Similar notations for the **outcome variable**:

$$Y_i = \begin{cases} 1 & \text{if student } i \text{ passed the standardized test} \\ 0 & \text{if student } i \text{ failed the standardized test} \end{cases}$$

- Note: i is a placeholder to refer to a generic unit (student in our context):
 - ex) Y_{42} is the outcome for the 42nd unit.

A Tale of Two Students (Redux)

	AI Assistant	Test Result
Student 1	Tutoring without AI assistant	Fail
Student 2	Tutoring with AI assistant	Pass

becomes...

i	T_i	Y_i
Student 1	0	0
Student 2	1	1

Causal Effects & Counterfactuals

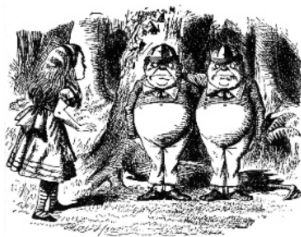
- What does “ T_i causes Y_i ” mean? $\rightsquigarrow$ **counterfactuals**, “what if”
- Would the student’s test result change with AI tutoring?
- Two **potential outcomes**:
 - $Y_i(1)$: would student i pass the test if s/he received AI-assisted tutoring?
 - $Y_i(0)$: would student i pass the test if s/he received tutoring w/o AI?
- **Causal effect**: $Y_i(1) - Y_i(0)$
 - $Y_i(1) - Y_i(0) = 0 \rightsquigarrow$ AI tutoring has no effect on test results
 - $Y_i(1) - Y_i(0) = -1 \rightsquigarrow$ AI tutoring worsens test results
 - $Y_i(1) - Y_i(0) = +1 \rightsquigarrow$ AI tutoring improves test results

Potential Outcomes

i	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
Student 1	0	0	???	0
Student 2	1	1	1	???

- **Fundamental problem of causal inference:**
 - We only observe one of the two potential outcomes.
 - Observe $Y_i = Y_i(1)$ if $T_i = 1$ or $Y_i = Y_i(0)$ if $T_i = 0$
- To infer causal effect, we need to infer the missing counterfactuals!

How to Figure Out Counterfactuals?



- Find a similar unit! $\rightsquigarrow$ **matching**
 - Mill's method of difference
- Did a student pass a standardized test because s/he had access to AI-assisted tutoring?
 - $\rightsquigarrow$ find an identical student who received tutoring w/o AI!
- New Jersey (NJ) increased the minimum wage. Causal effect on unemployment?
 - $\rightsquigarrow$ find a state similar to NJ that didn't increase minimum wage!

Imperfect Matches

ARNOLD SCHWARZENEGGER DANNY DEVITO
TWINS



Only their mothers can tell them apart...

- The problem: imperfect matches!
- Say we match i (treated) and j (control)
- **Selection Bias:** $Y_i(1) \neq Y_j(1)$
- Those who take treatment may be different than those who take control.
- Q: How can we correct for that?

Since (1) counterfactuals are never directly observable and (2) perfect matches are rarely attainable, we zoom out and analyze the problem at the aggregate level!

2/ Randomized experiments

Match Groups, Not Individuals!



- **Randomized control trials:** each unit's treatment assignment is determined by chance.
 - ex) flip a coin or draw red and blue chips from a hat, etc.
- Randomization ensures **balance** between treatment and control group.
 - Treatment and control group are identical **on average**
 - Similar on both observable and unobservable characteristics.

A Little More Notation

- We will often refer to the **sample size** (number of units) as n .
- We often have n measurements of some variable: $(Y_1, Y_2, \dots, Y_n)$
- Q: How many students in our sample data pass the standardized test?

$$Y_1 + Y_2 + Y_3 + \dots + Y_n$$

- Notation is a bit clunky/heavy, so we often use the **Sigma notation**:

$$\sum_{i=1}^n Y_i = Y_1 + Y_2 + Y_3 + \dots + Y_n$$

- $\sum_{i=1}^n$ means sum each value from Y_1 to Y_n

Averages

- The **sample average** or **sample mean** is simply the sum of all values divided by the number of values.
- Sigma notation allows us to write this in a compact way:

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

- Ex) Suppose we observed 6 students and found out 3 passed the test:

$$\bar{Y} = \frac{1}{6}(1 + 1 + 1 + 0 + 0 + 0) = 0.5$$

Quantity of interest

- We want to estimate the average causal effects over all units:

$$\begin{aligned}\text{Sample Average Treatment Effect (SATE)} &= \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\} \\ &= \frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n} \sum_{i=1}^n Y_i(0)\end{aligned}$$

- Q1: Why can't we just calculate this quantity directly?

Quantity of interest

- We want to estimate the average causal effects over all units:

$$\begin{aligned}\text{Sample Average Treatment Effect (SATE)} &= \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\} \\ &= \frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n} \sum_{i=1}^n Y_i(0)\end{aligned}$$

- Q1: Why can't we just calculate this quantity directly?
- What we can estimate instead:

$$\text{Difference in means} = \bar{Y}_{\text{treated}} - \bar{Y}_{\text{control}}$$

- $\bar{Y}_{\text{treated}}$: sample average outcome for treated group
- $\bar{Y}_{\text{control}}$: sample average outcome for control group
- Q2: When will the difference-in-means be a good estimate of the SATE?

Why Randomization Works

- Under an RCT, treatment and control groups are random samples.
- Average in the treatment group will be similar to average if all treated:

$$\bar{Y}_{\text{treated}} \approx \frac{1}{n} \sum_{i=1}^n Y_i(1)$$

- Average in the control group will be similar to average if all untreated:

$$\bar{Y}_{\text{control}} \approx \frac{1}{n} \sum_{i=1}^n Y_i(0)$$

- Implies difference-in-means should be close to SATE:

$$\bar{Y}_{\text{treated}} - \bar{Y}_{\text{control}} \approx \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\} = \frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n} \sum_{i=1}^n Y_i(0) = \text{SATE}$$

Balance Checking

- Can we determine if randomization “worked”?
- If it did, we shouldn’t see large differences between treatment and control group on **pretreatment variables**.
 - Pretreatment variables are those that are unaffected by treatment (e.g., does receiving AI tutoring affect gender? age?).
- We can check in the actual data for some pretreatment variable X :
 - $\bar{X}_{\text{treated}}$: average value of variable for treated group.
 - $\bar{X}_{\text{control}}$: average value of variable for control group.

⇒ Under randomization, $\bar{X}_{\text{treated}} - \bar{X}_{\text{control}} \approx 0$

Multiple Treatments

- Instead of 1 treatment, we might have multiple **treatment arms**:
 - Control condition
 - Treatment A
 - Treatment B
 - Treatment C, etc.
- In this case, we will look at multiple comparisons:
 - $\bar{Y}_{\text{treated, A}} - \bar{Y}_{\text{control}}$
 - $\bar{Y}_{\text{treated, B}} - \bar{Y}_{\text{control}}$
 - $\bar{Y}_{\text{treated, A}} - \bar{Y}_{\text{treated, B}}$
- If treatment arms are randomly assigned, these differences will be good estimators for each causal contrast.

3/ Calculating effects

AI Tutoring Study Data (Simulated)

```
1 # don't forget to load 'dplyr' and 'tidyverse'!  
2 # today's data are in the course data package pbadata (update it once: see the top of the R script)  
3 > library(pbadata)  
4 > AI_data <- ai_tutoring
```

Variable Name	Description
student_age	Age of the student in years
student_gender	Gender of the student (1 for Male, 0 for Female)
treat.ind	Treatment indicator (1 for treated with AI tutoring, 0 for control)
tutor_age	Age of the tutor in years
years_of_experience	Years of tutoring experience
education_level	Tutor's education level
test_outcome_pre	Pre-treatment test outcome (1 for Pass, 0 for Fail)
test_outcome_post	Post-treatment test outcome (1 for Pass, 0 for Fail)

- This is a 'simulated' experimental data we will examine to learn how to calculate the effects using R!
- The simulated data contains information on:
 - Student-level variables (e.g., students' age, gender)
 - Tutor-level variables (e.g., tutor's educational level, years of experience)

Peek at the Data

```
1 > AI_data # Take a peek at the data!
2
3 # A tibble: 500 × 8
4   treat_ind test_outcome_pre test_outcome_post student_age student_gender tutor_age years_of_experience
5     <int>         <int>         <int>         <int>         <int>         <int>         <int>
6 1         1             0             1            11             0            27            17
7 2         1             0             1             6             1            45            10
8 3         0             1             1            15             1            54            14
9 4         0             1             1             6             1            50            18
10 5         0             1             0             8             1            64            13
11 6         1             1             1            15             1            43            20
12 7         1             0             1             7             1            47             6
13 8         1             1             1            16             0            49            14
14 9         0             0             1            12             1            42            11
15 10        1             1             1            11             1            28            11
16 # i 490 more rows
17 # i 1 more variable: education_level <chr>
18 # i Use `print(n = ...)` to see more rows
```

- Again, the question was does AI tutoring help (on average)?

Compute the Ave. Outcomes in Each Group

- Average outcome for the treated group:

```
1 > treat_mean <- AI_data |>
2   filter(treat_ind == 1) |>
3   summarize(test_outcome_mean = mean(test_outcome_post))
4
5 > treat_mean
6 # A tibble: 1 × 1
7   test_outcome_mean
8         <dbl>
9 1             0.690
```

- Average outcome for the control group:

```
1 > control_mean <- AI_data |>
2   filter(treat_ind == 0) |>
3   summarize(test_outcome_mean = mean(test_outcome_post))
4
5 > control_mean
6 # A tibble: 1 × 1
7   test_outcome_mean
8         <dbl>
9 1             0.651
```

Calculate the Difference in Means

- Difference in means (treated minus control):

```
1 > treat_mean - control_mean
2
3   test_outcome_mean
4 1      0.03903557
```

- Again, this is a good estimate of the causal effect of AI tutoring if and **only if the two groups are statistically identical matches!**
- $\rightsquigarrow$ This would be the case if randomization “worked”.
- We’ll see more ways to do this throughout the semester.

Checking Balance on Numeric Covariates

- We can use `group_by()` to see how the mean of covariates varies by group:

```
1 > AI_data |>
2   group_by(treat_ind) |>
3   summarize(age_mean = mean(student_age))
4
5 # A tibble: 2 × 2
6   treat_ind age_mean
7     <int>     <dbl>
8 1         0      11.7
9 2         1      11.4
```

- Q: How can we do this for all numeric variables in the data? (hint: use `across()` and `where()`)
 - Post your codes on eclass! (voluntary)

Checking Balance on Categorical Covariates

- We can group by treatment and categorical control/covariate:

```
1 > AI_data |>
2   group_by(treat_ind, education_level) |>
3   summarize(n = n())
4
5 # A tibble: 9 × 3
6 # Groups:   treat_ind [2]
7   treat_ind education_level     n
8   <int>    <chr>          <int>
9 1         0 Bachelor's degree    52
10 2         0 High school         63
11 3         0 Master's degree     73
12 4         0 Two-year college    73
13 5         1 Bachelor's degree    65
14 6         1 Doctoral/PhD         4
15 7         1 High school         78
16 8         1 Master's degree     18
17 9         1 Two-year college    74
```

~> Hard to read!

pivot_wider()

- `pivot_wider()` takes data from a single column and moves it into multiple columns based on a grouping variable:

```
1 > AI_data |>
2   group_by(treat_ind, education_level) |>
3   summarize(n = n()) |>
4   pivot_wider(
5     names_from = treat_ind,
6     values_from = n
7   )
8
9 # A tibble: 5 × 3
10  education_level    `0`    `1`
11    <chr>          <int> <int>
12 1 Bachelor's degree    52    65
13 2 High school         63    78
14 3 Master's degree     73    18
15 4 Two-year college    73    74
16 5 Doctoral/PhD        NA     4
```

- `names_from` tells us what variable will map onto the columns.
- `values_from` tells us what values should be mapped into those columns.

Calculating Diff-in-Means by Group

- The mean differences could be more pronounced for some (sub)group of students compared to others, e.g.,:

```
1 > AI_data |>
2   mutate(
3     treat_ind = if_else(treat_ind == 1, "Treated", "Control"),
4     male_stu = if_else(student_gender == 1, "Male", "Female")
5   ) |>
6   group_by(treat_ind, male_stu) |>
7   summarize(test_outcome_mean = mean(test_outcome_post)) |>
8   pivot_wider(
9     names_from = treat_ind,
10    values_from = test_outcome_mean
11  ) |>
12  mutate(
13    diff_in_means = Treated - Control
14  )
15
16 # A tibble: 2 × 4
17   male_stu Control Treated diff_in_means
18   <chr>      <dbl>   <dbl>         <dbl>
19 1 Female    0.702    0.75         0.0475
20 2 Male     0.607    0.630         0.0231
```

- Q: Anything interesting that stands out to you?

Some Potential Problems with RCTs

- **Placebo effects:**

- Students experience a benefit from the tutoring solely because they believe AI tutoring is superior, even if the AI isn't actually adding any benefit.
- Q: How could we address this issue?

- **Hawthorne effects:**

- Students change their behavior (e.g., paying more attention, spending more time studying) because they know they are under study.

- Also, **practicality:**

- Can we run an experiment (and analyze the data) every time we want to study the effect of some managerial intervention (e.g., new type of advertising)?

4/ Observational studies

Context: Mobile Payment Study

- Data scientists observe ‘naturally assigned’ treatments (e.g., customers’ choice to adopt Line Pay).
 - As we discussed: often can’t randomize for ethical/logistical reasons.



- Can mobile payment services (e.g., Line Pay) change users’ shopping behavior on mobile commerce?
- Q: Why not just compare Line Pay adopters with non-adopters?
 - Post your thoughts on **eecclass**!

Line Pay Adoption Data (Simulated)

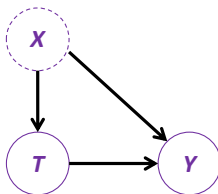
```
1 # don't forget to load 'dplyr' and 'tidyverse'!  
2 > LinePay_data <- linepay # also in the course data package, pbadata
```

Variable Name	Variable Description
line_pay_adopt	Indicator for Line Pay adoption (1 for Adopted, 0 for Not Adopted)
tech_savviness	Binary indicator for technological savviness (0 for Not Tech-Savvy, 1 for Tech-Savvy)
age	Customer's age, ranging from 18 to 65 years
gender	Customer's gender (0 for Male, 1 for Female)
occupation_type	Type of occupation (e.g., Student, Engineer, Doctor, Teacher, etc.)
spending_pre	Monthly M-commerce spending amount in USD before the launch of Line Pay in the early 2020
spending_post	Monthly M-commerce spending amount in USD after the launch of Line Pay in the early 2020

Peek at the Data

```
1 > LinePay_data
2
3 # A tibble: 500 × 7
4   line_pay_adopt tech_savviness age gender occupation_type spending_pre spending_post
5   <int>          <int> <int> <int> <chr>          <dbl>      <dbl>
6 1             0         1  25    1 Teacher        51.2      63.0
7 2             0         1  47    0 Teacher        73.3      84.7
8 3             1         0  62    1 Teacher        48.5      72.6
9 4             0         0  47    0 Teacher        18.2      30.0
10 5             0         0  23    0 Student        62.9      72.3
11 6             0         0  58    1 Other          59.6      67.8
12 7             1         1  31    1 Teacher        65.1      89.7
13 8             1         0  33    1 Teacher        49.6      76.9
14 9             1         0  37    0 Engineer        42.7      67.3
15 10            0         1  36    0 Doctor          50.1      56.4
16 # i 490 more rows
17 # i Use `print(n = ...)` to see more rows
```

Confounding



- **Confounder:** pretreatment variable affecting both the treatment & the outcome.
 - Tech-savvy customers (X) are more likely to adopt Line Pay (T).
 - Tech-savvy customers (X) are also more likely to purchase from m-commerce (Y).
- **Confounding bias** in the estimated SATE due to these differences
 - $\bar{Y}_{\text{control}}$ not a good proxy for $\frac{1}{n} \sum_{i=1}^n Y_i(0)$ in treated group.

Research Designs

- Q: How Do We Address This Issue?
- Q. restated: How can we find a good comparison group?
 - Depends on the data we have available.
- Three general ways to analyze observational data:
 1. **Cross-sectional design:** compare outcomes of treated and control units at one point in time.
 2. **Before-and-after design:** compare outcomes before and after a unit has been treated, but need over-time data on treated group.
 3. **Difference-in-differences design:** use before/after information for the treated and control group; need over-time data on treated & control groups.

Cross-Sectional Design

- Compare Line Pay adopters (treated) and non-adopters (control) after the launch of Line Pay in the early 2020.
 - Customers who adopted Line Pay vs. those who did not.
- Adopters and non-adopters assumed identical on average as in RCT.
- Cross-section comparison estimate:

$$\overline{Y}_{\text{after}}^{\text{treated}} - \overline{Y}_{\text{after}}^{\text{control}}$$

- Could there be confounders? (e.g., tech-savviness)

Cross-Sectional Design in R

- Sample average outcome for treated & control after adoption:

```
1 > adopted <- LinePay_data |>
2   filter(line_pay_adopt == 1) |>
3   summarize(mean(spending_post)); adopted
4
5 # A tibble: 1 × 1
6   `mean(spending_post)`
7   <dbl>
8 1          82.7
9
10 > no_change <- LinePay_data |>
11   filter(line_pay_adopt == 0) |>
12   summarize(mean(spending_post)); no_change
13
14 # A tibble: 1 × 1
15   `mean(spending_post)`
16   <dbl>
17 1          59.3
```

- Take the difference!:

```
1 > adopted - no_change
2
3   mean(spending_post)
4 1          23.37544
```

Statistical Control

- **Statistical control:** adjust for confounders using statistical procedures.
 - Can help to reduce confounding bias.
- One type of statistical control: **subclassification**
 - Compare treated and control groups within levels of a confounder.
 - e.g., compare adopters with non-adopters among tech-savvy customers, and again among customers who are NOT tech-savvy.
 - Remaining effect can't be due to the confounder.

Statistical Control in R

```
1 > LinePay_data |>
2   group_by(tech_savviness, line_pay_adopt) |>
3   summarize(avg_spending = mean(spending_post)) |>
4   mutate(line_pay_adopt = if_else(line_pay_adopt == 1, "adopted", "unadopted")) |>
5   pivot_wider(
6     names_from = line_pay_adopt,
7     values_from = avg_spending
8   ) |>
9   mutate(diff_by_tech_savviness = `adopted` - `unadopted`)
10
11 # A tibble: 2 × 4
12 # Groups:   tech_savviness [2]
13   tech_savviness unadopted adopted diff_by_tech_savviness
14   <int>          <dbl>    <dbl>                <dbl>
15 1         0         53.7     73.8                 20.1
16 2         1         69.0     88.0                 19.0
```

- Limitation: we can only control for observed variables $\rightsquigarrow$ threat from **‘unmeasured’ confounding**

Before-and-After Comparison

- Compare m-commerce spending of Line Pay adopters before & after the launch of Line Pay in the early 2020.
- Advantage: all user-specific features held fixed
 - comparing within a user over time.
- Before-and-after estimate:

$$\overline{Y}_{\text{adopters}}^{\text{after}} - \overline{Y}_{\text{adopters}}^{\text{before}}$$

Before and After Design in R

```
1 > LinePay_data |>
2   filter(line_pay_adopt == 1) |>
3   mutate(
4     spending_change = spending_post - spending_pre
5   ) |>
6   summarize(avg_change = mean(spending_change))
7
8 # A tibble: 1 × 1
9   avg_change
10   <dbl>
11 1      28.4
```

- Limitation: **time-varying confounders**
 - If shopping paradigm was simply moving to mobile commerce (with more people shopping on their phones).

Differences in Differences

- Key idea: use the before-and-after difference of **control group** to infer what would have happened to **treated group** without treatment.
- DiD estimate:

$$\underbrace{(\bar{Y}_{\text{treated}}^{\text{after}} - \bar{Y}_{\text{treated}}^{\text{before}})}_{\text{trend in treated group}} - \underbrace{(\bar{Y}_{\text{control}}^{\text{after}} - \bar{Y}_{\text{control}}^{\text{before}})}_{\text{trend in control group}}$$

- Change in treated group above and beyond the change in control group.
- **Parallel time trend assumption**
 - Changes in m-commerce spending for Line Pay adopters would be roughly the same as non-adopters if Line Pay wasn't adopted.
 - Limitations: non-parallel trends—what if adopters and non-adopters had different time trends?

Difference-in-Differences in R

```
1 > LinePay_data |>
2   mutate(
3     spending_change = spending_post - spending_pre,
4     adopted = if_else(line_pay_adopt == 1, "adopted", "unadopted")
5   ) |>
6   group_by(adopted) |>
7   summarize(avg_change = mean(spending_change)) |>
8   pivot_wider(
9     names_from = adopted,
10    values_from = avg_change
11  ) |>
12   mutate(DID = adopted - unadopted)
13
14 # A tibble: 1 × 3
15   adopted unadopted   DID
16   <dbl>     <dbl> <dbl>
17 1    28.4        8.03  20.4
```

Summarizing Approaches

1. **Cross-sectional comparison**

- Compare treated units with control units after treatment
- Assumption: treated and control units are comparable
- Possible confounding

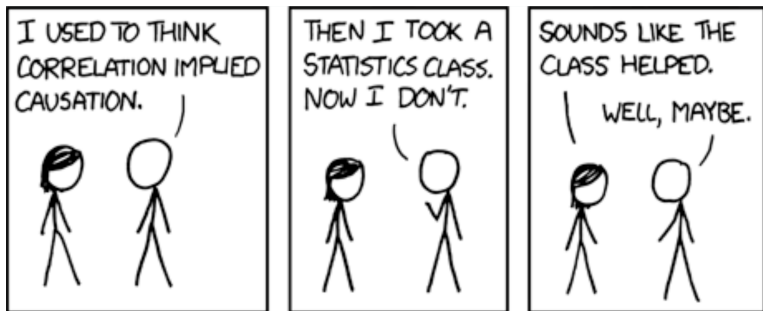
2. **Before-and-after comparison**

- Compare the same units before and after treatment
- Assumption: no time-varying confounding

3. **Differences-in-differences**

- Assumption: parallel trends
 - Under this assumption, it accounts for unit-specific and time-varying confounding.
-
- All rely on assumptions that aren't easily verified.
 - RCTs (experiments) handle confounding by design.

Causality Understanding Check



Source: xkcd.com/552

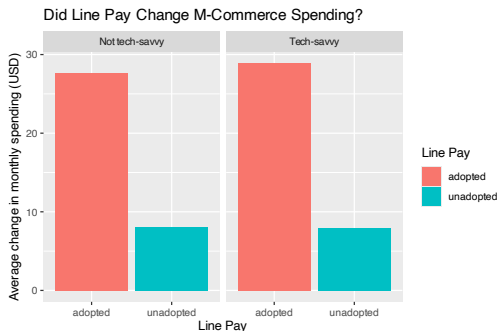
5/ In-Class Exercise E3

In-Class Exercises: Background

- We want to find out the answer to the following question:
 - Q: did adopting Line Pay change customers' m-commerce spending?
- The LinePay_data from today's class can answer this, so keep working in the same R session.
 - Some relevant variables:
 - line_pay_adopt: 1 if the customer adopted Line Pay, 0 otherwise
 - tech_savviness: 1 if the customer is tech-savvy, 0 otherwise
 - spending_pre, spending_post: monthly m-commerce spending (USD) before and after the launch of Line Pay

In-Class Exercises: Task

- Recreate the following plot:



- The plot shows the average change in monthly m-commerce spending (after minus before) for adopters and non-adopters, separately for customers who are not tech-savvy and customers who are.
 - To get started, we'll give you a few clues on the following slide.

In-Class Exercises: Instructions

1. Start from the difference-in-differences code on slide 46.
2. In the same `mutate()`, create `tech` with `if_else()`: "Tech-savvy" when `tech_savviness == 1`, otherwise "Not tech-savvy".
3. `group_by(tech, adopted)`, then `summarize()` two things: the mean of `spending_change` and the number of customers (`n = n()`). Your table has 4 rows, and `n` adds up to 500.
4. Plot with `geom_col()`: `adopted` on x, your average on y, `fill = adopted`, and one panel per tech with `facet_wrap()` (Week 2). Add `labs(title = , x = , y = , fill = )`.
5. Compute the difference in differences within each panel: drop `n` with `select(-n)`, then use `pivot_wider()` as on slide 46. Verify before you trust the output: you should get 2 rows and no NA.

Q: Is the effect of Line Pay different for tech-savvy customers? Compare your two estimates with each other and with the estimate for all customers on slide 46 (20.4). Look at `n`: how sure are you about the gap?

What Comes Next?

- Today: what a causal effect is, why randomized experiments work, and three designs for observational data.
- Next week (Oct 8): **causality II**. DAGs and covariate selection: what to control for, and what not to, in your project. Review IMS §2.3 beforehand.
- **Assignment 1 is due before class starts next Thursday, Oct 8 (graduate).** Assignment 2 is released that day.
- Milestone 1 (GitHub repository) is due Sunday, Oct 11.
- **E3 (today's exercise): submit on eeclass by 23:59 today.**

Enjoy your Weekends! :)

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